ShadowMem

Automated Volatile Memory Forensics & Fileless Malware Detection

Platform: Windows 10/11 x64 | **Backend:** FastAPI & Python 3.13 | **Frontend:** Next.js 14 SOC | **Generated:** October 31, 2026

1. Executive Summary & Problem Space

Modern threat actors, advanced persistent threats (APTs), and ransomware operators increasingly employ **fileless attack techniques**—operating entirely inside system RAM without writing a single binary to disk. Conventional signature-based endpoint detection mechanisms that rely on file hashes and disk traversal remain blind to these techniques. ShadowMem bridges this security chasm by directly introspecting 64-bit Windows process Virtual Address Descriptors (VAD), thread start addresses, and memory protection transitions in real-time.

2. System Architecture & Dual-Tier Design

ShadowMem is architected as an authoritative, dual-tier platform engineered for speed, low overhead, and forensic rigor:

- **Authoritative FastAPI Kernel:** Exposes RESTful forensic endpoints, real-time WebSocket feeds, and Win32 ctypes memory bindings.
- **Compiled Next.js 14 Dashboard:** Pre-rendered to pure static assets, hosted directly by FastAPI on port 8000 with zero separate Node.js server required at runtime.
- **Parallel Introspection Engine:** Thread-pooled VirtualQueryEx traversal scanning 150+ processes across all user-space memory in ~21 seconds.

Component	Technology / Module	Key Role in Forensic Pipeline
VAD Walker	memoryscanner.py (Win32 ctypes)	Traverses process virtual address space via VirtualQueryEx; detects unbacked private executable allocations.
Thread Inspector	ntdll.dll (ThreadQuerySetWin32StartAddress)	Queries thread instruction entry points; detects execution spawned in dynamic memory buffers.
Classifier Pipeline	services/forensics.py & ML model	Evaluates 7-dimensional forensic telemetry vector; applies JIT compiler false-positive suppression.
Explainable AI (XAI)	Feature Attribution Engine	Calculates percentage contribution of each feature to threat score with forensic rationale.
Memory Hex Dumper	ReadProcessMemory API	Extracts 256-byte slice of suspect memory; computes Shannon entropy and detects shellcode preambles.
Surgical Containment	services/containment.py	Executes thread-level suspension via SuspendThread without crashing system stability.

3. Low-Level Win32 Memory Forensics Engine

The memory inspection subsystem directly queries the Windows kernel subsystem via Win32 C APIs:

• **Unbacked Executable Memory:** When memory allocations have MEM_COMMIT + MEM_PRIVATE with execution privileges (PAGE_EXECUTE_READWRITE or PAGE_EXECUTE_READ), they lack backing from an on-disk image file. This is the primary indicator of staged payloads and reflective DLLs.

• **Protection Transition Anomaly:** Malware commonly allocates PAGE_READWRITE memory, copies encrypted shellcode, then flips the region to executable via VirtualProtect. ShadowMem flags allocations where AllocationProtect == PAGE_READWRITE but Protect is executable.

• **Thread Start Verification:** Queries NtQueryInformationThread class 9. If a thread's start address resolves within an unbacked dynamic memory page, execution hijacking is confirmed.

• **Shannon Entropy Slicing:** Safely reads 256 bytes via ReadProcessMemory and computes entropy $H(X)$. Values approaching 8.0 confirm encrypted or compressed shellcode (e.g. Cobalt Strike beacon headers).

4. 7-Feature ML Model & Explainable AI (XAI)

ShadowMem constructs a 7-dimensional forensic telemetry vector per process for inference:

Feature	Type	Forensic Description	Adversarial Signal
rwx_count	Integer	Number of PAGE_EXECUTE_READWRITE committed memory allocations	Permissive code+data buffer
unbacked_exec_count	Integer	Count of private executable memory allocations without disk backing	Injected shellcode buffer
total_rwx_bytes	Integer	Aggregate byte volume mapped with RWX memory permissions	Payload size indicator
num_threads	Integer	Total active threads belonging to the target process	Process activity baseline
thread_unbacked_start	Integer	Threads whose initial Win32 start address resides in unbacked memory	Thread execution hijack
is_jit_flag	Float (0/1)	Indicator for legitimate JIT engines (Chrome, Edge, .NET CLR, Electron)	False-positive dampening
vad_protection_anomaly	Integer	Allocations created as PAGE_READWRITE and flipped to executable	Self-injection transition

5. MITRE ATT&CK; Adversary Technique Mapping

MITRE ID	Technique Name	Forensic Indicators Observed	Recommended Response
T1055.001	DLL Injection	Unlinked library in PEB, foreign thread creation in target process.	Suspend process threads; isolate host.
T1055.002	Reflective Code Loading	DOS 'MZ' header (0x4D 0x5A) detected in unbacked MEM_PRIVATE memory.	Freeze threads via NtSuspendProcess; dump hex.
T1055.003	Thread Execution Hijacking	Thread entry point residing directly inside unbacked executable buffer.	Immediate thread-level suspension.

T1055.012	Process Hollowing	Process PE header entry point modified; .text virtual size mismatch.	Terminate hollowed process; inspect parent.

6. Next.js 14 SOC Dashboard & Generative UI

The SOC Cyber Defense Dashboard features a refined Warm SOC aesthetic engineered for high-alert clarity:

- **Threat Posture Radial Ring:** SVG circular progress indicator computing real-time clearance rate with color-coded posture metrics.
- **Live Process Allocation Grid:** Interactive spatial map with category filters (ALL, THREATS, JIT, CLEAN) and real-time VAD telemetry tooltips.
- **Process Forensic Dossier Modal:** 3-tab deep-dive investigation interface: (1) Explainable AI Feature Attribution, (2) VAD Memory Regions breakdown, and (3) Live Hex & ASCII Disassembly Inspector with Shannon entropy gauge.
- **Streaming Activity Terminal:** Monospaced kernel activity feed with auto-scroll and WebSocket-driven telemetry frame logging.

7. Evaluation & Judge Presentation Playbook

When presenting ShadowMem to hackathon judges or technical evaluators, execute this 4-step sequence:

Step 1 — Real Workstation Audit: Click *'Run Volatile Scan'*. Demonstrate how 150+ live Windows processes and their thread contexts are evaluated via low-level Win32 calls in ~21s.

Step 2 — Live Attack Demo: Click *'Live Attack Demo'*. A simulated Cobalt Strike reflective injection alert broadcasts over WebSocket, elevating the posture meter and opening the dossier.

Step 3 — Forensic Proof (Hex Inspector): Switch to the *'Live Hex & Buffer Disassembly'* tab in the dossier. Show judges the 16-byte raw hex rows, ASCII decode, and 8.0/8.0 Shannon entropy score.

Step 4 — Active Containment: Click *'Isolate Target Process'*. Demonstrate surgical Win32 user-mode thread suspension and verify the audit trail in the Activity Feed.